

PMA Spark / Snowflake Secrets — Reference Guide

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Overview

This document lists all Secrets Manager containers created for PMA Spark/Airflow2 Snowflake access, across all AWS and GCP environments, along with the AWS EMR EC2 IAM role and GCP Dataproc service account access that was granted to read them. Four logical secrets exist in every environment:

- **pma_stat_snowflake_credentials_<tier>** — shared Snowflake stats credentials (JSON), same value copied from a single global source secret into every environment.
- **pma_snowflake_key_<tier>** — shared Snowflake private key, same value copied from a single global source secret into every environment.
- **pma_stat_sf_user** — per-environment Snowflake stats username (differs: production vs staging tiers).
- **pma_spark_sf_account** — per-environment Snowflake account identifier (differs per environment/region), sourced from the txt2sql SNOWFLAKE_ACCOUNT_MAPPING.

<tier> is **production** for all 12 production environments, and **staging** for aws-staging / gcp-staging.

AWS Secrets (AWS Secrets Manager)

AWS secrets are scoped uniquely by path prefix `mist/<env>/airflow2/<secret_name>`. iac terragrunt directory: `<env>/auto/secrets/<secret_name>`.

iac Environment	Secret Manager Path
aws-production	mist/production/airflow2/pma_stat_snowflake_credentials_production mist/production/airflow2/pma_snowflake_key_production mist/production/airflow2/pma_stat_sf_user mist/production/airflow2/pma_spark_sf_account
use1prod2-aws	mist/use1prod2/airflow2/pma_stat_snowflake_credentials_production mist/use1prod2/airflow2/pma_snowflake_key_production mist/use1prod2/airflow2/pma_stat_sf_user mist/use1prod2/airflow2/pma_spark_sf_account
apse2prod5-aws	mist/apse2prod5/airflow2/pma_stat_snowflake_credentials_production mist/apse2prod5/airflow2/pma_snowflake_key_production mist/apse2prod5/airflow2/pma_stat_sf_user mist/apse2prod5/airflow2/pma_spark_sf_account
eu-aws	mist/eu/airflow2/pma_stat_snowflake_credentials_production mist/eu/airflow2/pma_snowflake_key_production mist/eu/airflow2/pma_stat_sf_user mist/eu/airflow2/pma_spark_sf_account
aws-staging	mist/staging/airflow2/pma_stat_snowflake_credentials_staging mist/staging/airflow2/pma_snowflake_key_staging mist/staging/airflow2/pma_stat_sf_user mist/staging/airflow2/pma_spark_sf_account

GCP Secrets (GCP Secret Manager)

GCP secrets are scoped by project (the per-environment Dataproc project). Secret name is the same across all GCP environments; only the project differs. iac terragrunt directory: <env>/auto/secrets/<secret_name>.

iac Environment	Dataproc Project	Secret Names
gcp-production	mist-dataproc-prod	pma_stat_snowflake_credentials_production pma_snowflake_key_production pma_stat_sf_user pma_spark_sf_account
asne1prod7-gcp	mist-dataproc-asne1prod7	pma_stat_snowflake_credentials_production pma_snowflake_key_production pma_stat_sf_user pma_spark_sf_account
aso1prod5-gcp	mist-dataproc-aso1prod5	pma_stat_snowflake_credentials_production pma_snowflake_key_production pma_stat_sf_user pma_spark_sf_account
euwe1prod3-gcp	mist-dataproc-euwe1prod3	pma_stat_snowflake_credentials_production pma_snowflake_key_production pma_stat_sf_user pma_spark_sf_account
mec1prod6-gcp	mist-dataproc-mec1prod6	pma_stat_snowflake_credentials_production pma_snowflake_key_production pma_stat_sf_user pma_spark_sf_account
mec2prod6-gcp	mist-dataproc-mec2prod6	pma_stat_snowflake_credentials_production pma_snowflake_key_production pma_stat_sf_user pma_spark_sf_account
usc1prod4-gcp	mist-dataproc-usc1prod4	pma_stat_snowflake_credentials_production pma_snowflake_key_production pma_stat_sf_user pma_spark_sf_account
nane1prod1-gcp	mist-dataproc-nane1prod1	pma_stat_snowflake_credentials_production pma_snowflake_key_production pma_stat_sf_user pma_spark_sf_account
gcp-staging	mist-dataproc-staging	pma_stat_snowflake_credentials_staging pma_snowflake_key_staging pma_stat_sf_user pma_spark_sf_account

Runtime Access Granted

AWS — EMR EC2 IAM Role

All 5 AWS environments' EMR EC2 IAM role (via the shared terraform-aws-coreinfra iam/emr-v2-ec2 module, v2.5.11) has an AllowPmaSecretsAccess statement granting secretsmanager:GetSecretValue on all 4 secrets listed above for that environment's tier. aws-staging's previous inline statement (which had a naming bug and never worked) was removed and replaced by this shared module grant.

GCP — Dataproc Service Account

All 9 GCP environments' Dataproc service account already has project-wide roles/secretmanager.SecretAccessor, covering all 4 secrets automatically (6 environments manage this via Terraform; gcp-production and nane1prod1-gcp have it granted out-of-band, confirmed live via gcloud). No GCP changes were required.

Populating / Rotating Secret Values

- Shared secrets (pma_stat_snowflake_credentials_*, pma_snowflake_key_*) are populated via scripts/copy_global_pma_secret.sh in mistsys/iac (copies byte-exact from a single global source secret).
- Per-environment secrets (pma_stat_sf_user, pma_spark_sf_account) are populated via scripts/set_pma_secret_value.sh in mistsys/iac (explicit --value argument, supports --target=aws|gcp). Both scripts are idempotent and byte-exact verified.

All Pull Requests Merged for This Effort (mistsys/iac)

PR	Title
#4107	provision pma_stat_snowflake secret resources and EMR/Dataproc IAM
#4108	aws-production: migrate EMR and EMR-EC2 IAM roles from landcode to iac
#4214	aws-staging: migrate EMR IAM roles from landcode to iac
#4215	eu-aws: migrate EMR IAM roles from landcode to iac
#4216	use1prod2-aws: migrate EMR IAM roles from landcode to iac
#4235	apse2prod5: migrate EMR IAM to coreinfra modules
#4239	use1prod2: upgrade EMR IAM coreinfra modules to v2.5.8
#4240	aws-production: upgrade EMR IAM coreinfra modules to v2.5.8
#4778	aws-production: create pma_spark_stat_snowflake secret containers
#4779	use1prod2-aws: create pma_spark_stat_snowflake secret containers
#4780	apse2prod5-aws: create pma_spark_stat_snowflake secret containers
#4781	eu-aws: create pma_spark_stat_snowflake secret containers
#4782	aws-staging: create pma_spark_stat_snowflake secret containers
#4785	gcp-production: create pma_spark_stat_snowflake secret containers in dataproc project
#4786	asne1prod7-gcp: create pma_spark_stat_snowflake secret containers in dataproc project
#4787	aso1prod5-gcp: create pma_spark_stat_snowflake secret containers in dataproc project
#4788	euwe1prod3-gcp: create pma_spark_stat_snowflake secret containers in dataproc project

PR	Title
#4789	mec1prod6-gcp: create pma_spark_stat_snowflake secret containers in dataproc project
#4790	mec2prod6-gcp: create pma_spark_stat_snowflake secret containers in dataproc project
#4791	usc1prod4-gcp: create pma_spark_stat_snowflake secret containers in dataproc project
#4792	nane1prod1-gcp: create pma_spark_stat_snowflake secret containers in dataproc project
#4793	gcp-staging: create pma_spark_stat_snowflake secret containers in dataproc project
#4837	fix AWS credential precedence and validate fetched secret value in copy_global_pma_secret.sh
#4845	aws-staging: rename pma_spark_stat_snowflake_* secrets to match global naming
#4851	aws-production: create pma_stat_sf_user / pma_spark_sf_account secret containers
#4852	use1prod2-aws: create pma_stat_sf_user / pma_spark_sf_account secret containers
#4853	apse2prod5-aws: create pma_stat_sf_user / pma_spark_sf_account secret containers
#4854	eu-aws: create pma_stat_sf_user / pma_spark_sf_account secret containers
#4855	aws-staging: create pma_stat_sf_user / pma_spark_sf_account secret containers + set_pma_secret_value.sh
#4856	gcp-production: create pma_stat_sf_user / pma_spark_sf_account secret containers
#4857	asne1prod7-gcp: create pma_stat_sf_user / pma_spark_sf_account secret containers
#4858	aso1prod5-gcp: create pma_stat_sf_user / pma_spark_sf_account secret containers
#4859	euwe1prod3-gcp: create pma_stat_sf_user / pma_spark_sf_account secret containers
#4860	mec1prod6-gcp: create pma_stat_sf_user / pma_spark_sf_account secret containers
#4861	mec2prod6-gcp: create pma_stat_sf_user / pma_spark_sf_account secret containers
#4862	usc1prod4-gcp: create pma_stat_sf_user / pma_spark_sf_account secret containers
#4863	nane1prod1-gcp: create pma_stat_sf_user / pma_spark_sf_account secret containers
#4864	gcp-staging: create pma_stat_sf_user / pma_spark_sf_account secret containers + set_pma_secret_value.sh
#4865	aws-production: grant EMR EC2 role access to PMA Snowflake secrets
#4866	use1prod2-aws: grant EMR EC2 role access to PMA Snowflake secrets
#4867	apse2prod5-aws: grant EMR EC2 role access to PMA Snowflake secrets
#4868	eu-aws: grant EMR EC2 role access to PMA Snowflake secrets

Related PR (mistsys/terraform-aws-coreinfra)

- #209 — grant EMR EC2 role access to PMA Snowflake secrets (shared module, merged and tagged v2.5.11).

Pending at Time of Writing

- iac#4869 (aws-staging: grant EMR EC2 role access to PMA Snowflake secrets) — open, pending review/merge.